

ABTR case

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1 ABTR description

A demonstrative system model based on the Advanced Burner Test Reactor (ABTR) conceptual design has been developed with the SAM for the reactor core.

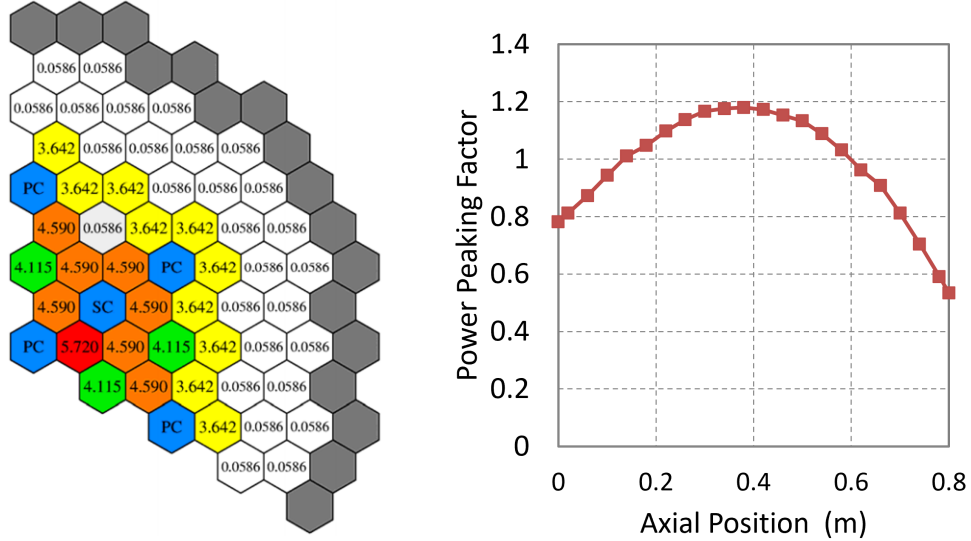


Figure 1: Assembly radial (left) and axial (right) power distribution at BOC.

The primary system is configured in a pool-type arrangement, with the reactor core. The primary coolant is sodium. The reactor core consists of 24 assemblies in an inner enrichment zone and 30 assemblies in the outer zone. A total of nine test locations are provided for fuel (6 assemblies) and material (3 assemblies) tests. On the basis of the reactor physics calculations, a five-channel model was selected to model the reactor core. Channel 1 is used to represent the peak-power inner-core subassembly with fresh fuel. Channels 2 and 4 represent the average subassemblies in the inner and outer enrichment zones respectively, while channel 3 represents the average of the mid-core fuel test assemblies. Channel 5 represents all of the non-fuel subassemblies, including the mid-core materials test assemblies. Figure 1 shows the initial subassembly powers at the beginning of equilibrium cycle conditions, and the average axial power shape for all assemblies. The geometric data and input conditions employed in the multi-channel core model are shown in Table 1.

Table 2 shows additional characteristics of the core channels including friction factor, heat transfer coefficient, materials of heat structure, power fraction, and heat transfer area density.

The material properties used in this model include four solid materials: **fuel-mat**, **gap-mat**, **clad-mat**, and **duct-mat**. The fuel-mat is defined with a thermal conductivity of 16 W/m·K, a heat capacity (C_p) of 191.67 J/kg·K, and a density of 1.4583×10^4 kg/m³. The gap-mat has a thermal conductivity of 64 W/m·K, a heat capacity of 1272 J/kg·K, and a density of 865 kg/m³. The clad-mat is modeled with a thermal

Table 1: Component Geometry Summary

Component	A (m ²)	Dh (m)	Length (m)
CH1	4.9237e-3	2.972e-3	0.8
CH2	0.11323	2.972e-3	0.8
CH3	0.029539	2.972e-3	0.8
CH4	0.14769	2.972e-3	0.8
CH5	0.153955129	2.972e-3	0.8
pipe1	0.44934	2.972e-3	0.6
pipe2	0.44934	2.972e-3	1.5

Table 2: Heat Structure, Thermal Parameters, and Power Fraction

Component	n_{hs}	Width of HS (m)	Material	f	Hw	HT Area Density	Power Fraction
CH1	3	0.003015/0.000465/0.00052	fuel/gap/clad	0.022	1.6129e5	1107.8	0.02248
CH2	3	0.003015/0.000465/0.00052	fuel/gap/clad	0.022	1.6129e5	1107.8	0.41924
CH3	3	0.003015/0.000465/0.00052	fuel/gap/clad	0.022	1.6129e5	1107.8	0.09852
CH4	3	0.003015/0.000465/0.00052	fuel/gap/clad	0.022	1.6129e5	1107.8	0.43116
CH5	2	6.32340e-3/7.0260e-4	fuel/clad	0.04	13619	2113.6	0.02860

conductivity of 26 W/m·K, a heat capacity of 638 J/kg·K, and a density of 7.646×10^3 kg/m³. The duct-mat uses the same thermophysical properties as the cladding, with $k = 26$ W/m·K, $C_p = 638$ J/kg·K, and $\rho = 7.646 \times 10^3$ kg/m³. These values define the thermal response of all solid regions in the heat-structure system.

The thermal-hydraulic model consists of five parallel fuel channels (CH1–CH5) that are all vertically oriented along the positive z -axis and share identical inlet and outlet flow directions. These channels represent parallel flow paths within the core. The parallel channels are hydraulically connected between two one-dimensional pipes that serve as the inlet and outlet plenums of the system. Upstream of the channels, **pipe1** delivers coolant to a branching junction, which distributes the flow into all five channels simultaneously. This junction includes individual form-loss coefficients (K values) to represent flow-distribution resistances for each channel leg. Specifically, the values used in **branch1** are $K = 0.1, 0.5, 1.96, 2.16, 4.5$, and 3500.0, corresponding respectively to the main pipe passage and the five channel inlets (CH1–CH5). Downstream, the outlets of all five channels recombine at a second junction (**branch2**), where the form-loss coefficients are all set to zero ($K = 0.0$ for each outlet leg), allowing the flows to merge without additional hydraulic resistance. This junction gathers the returning coolant and directs it into **pipe2**, completing the loop back toward the outlet boundary. The branches area is 0.44934.

The boundary conditions are imposed using two TDV components: **inlet_TDV** specifies the inlet velocity and temperature, enforcing a velocity boundary condition of 3.25 m/s and an inlet coolant temperature of 628.15 K at the entrance of **pipe1**. At the system outlet, **outlet_TDV** applies a fixed pressure of 1×10^5 Pa along with a temperature boundary condition of 628.15 K at the exit of **pipe2**. Together, these boundaries define the global driving conditions for the transient simulation, while the internal distribution of mass flow among channels is governed by the hydraulic resistances and form losses specified within the branching components.

The initial conditions for the simulation are defined globally, setting the thermodynamic state of the system before the transient begins. The coolant throughout all fluid components is initialized at a uniform pressure of 1.0×10^5 Pa, a velocity of 2 m/s, and a temperature of 628.15 K. These global values establish a consistent starting point for all channels, pipes, and junctions. For the solid regions of the heat structures, the initial temperature is also set to 628.15 K, ensuring thermal equilibrium between the solid and fluid domains at startup. A temperature scaling factor of $T_{\text{solid_sf}} = 1 \times 10^{-3}$ and variable scaling factors of 1, 10^{-3} , and 10^{-6} are applied to improve numerical conditioning during the nonlinear solve. Together, these initial conditions provide a well-defined, steady baseline from which the transient evolution of mass flow, temperature, and heat transfer can develop.

The numerical discretization of the system is defined through both axial and horizontal resolutions. Vertically, each core channel (CH1–CH5) is modeled using 20 axial fluid elements over a heated length of

0.8 m, providing uniform spatial resolution for capturing coolant temperature rise, pressure drop, and heat transfer variations along the flow direction. The heat structures associated with each channel are likewise discretized axially according to the specified element counts for the fuel, gap, and cladding layers (5, 1, and 1 elements, respectively, for the reference fuel channels; 5 and 1 elements for the reflector channel). Horizontally, the model resolves five parallel channels coupled through their inlet and outlet junctions.

The time discretization for this case is governed by a transient executioner combined with a piecewise linear time-step control function. Although the physical boundary conditions remain constant throughout the simulation, the transient framework allows the solution to evolve in time toward a numerically converged state. The base time step is initially very small, with values defined by the `time_stepper` function: $\Delta t = 0.01$ s from $t = 0$ to 0.1 s, increasing to $\Delta t = 0.1$ s by $t = 0.2$ s, and eventually reaching a maximum allowable step size of $\Delta t = 1$ s for times beyond 2 s. This gradual increase in time-step size ensures stability and accuracy during the early stages of the computation, where strong nonlinear coupling between fluid and solid temperatures may require tighter control. The executioner further enforces limits of $\Delta t_{\min} = 10^{-5}$ s and a nominal starting step of $\Delta t = 0.1$ s, allowing the nonlinear solver to reduce the time step adaptively as needed. Despite the transient formulation, only five time steps are executed, making the simulation effectively a pseudo-steady-state solve in which the time discretization primarily serves as a numerical mechanism to achieve convergence rather than to resolve physical temporal behavior.